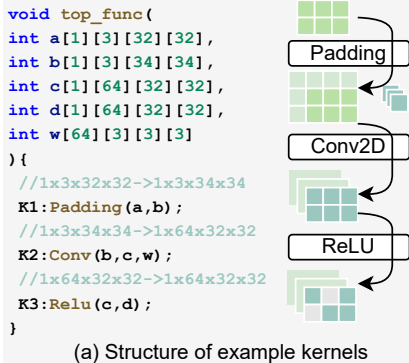
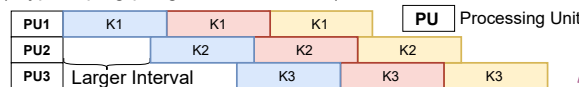




Framework	Input Size	Latency/Cycles↓	Speedup↑
Baseline-1	(1, 3, 32, 32)	12538920	1.00x
ScaleHLS	(1, 3, 32, 32)	886342	14.15x
POM	(1, 3, 32, 32)	372771	33.64x
Allo	(1, 3, 32, 32)	221964 (Deadlock)	56.49x
StreamHLS	(1, 3, 32, 32)	224479	55.86x
CODO	(1, 3, 32, 32)	102886	121.87x
Baseline-2	(1, 3, 224, 224)	3089907273	1.00x
HIDA	(1, 3, 224, 224)	4019817	768.67x
CODO	(1, 3, 224, 224)	1692801	1825.32x

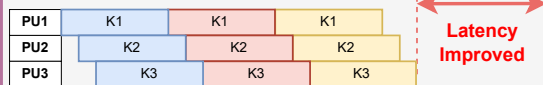
(b) Evaluation results on example kernels

(c) Typical ping-pong-based dataflow (ScaleHLS, POM, and HIDA)

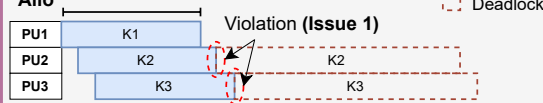


(d) Existing FIFO-based dataflow (Allo, StreamHLS, and StreamTensor)

Ideal execution



Allo Longer Latency (Issue 2 ①)



StreamHLS and StreamTensor Sequential / Ping-pong-based Execution (Issue 1)



(e) Optimized FIFO-based dataflow (CODO)

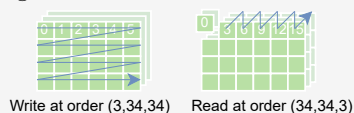


Issue 1 Dataflow Violation

// Padding

for ci=0...3
for h=0...34
for w=0...34
write(FIFO)

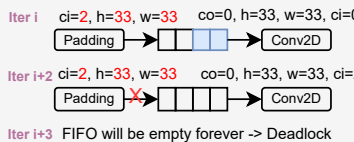
① Access order mismatch



// Conv2D

for co=0...64
for h=0...34
for w=0...34
for ci=0...3
read(FIFO)

② Access count inconsistency

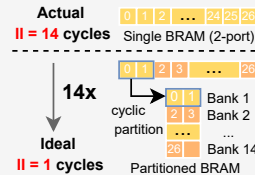


Issue 2 Insufficient FIFO optimization

① Restricted array partition

// Conv2D

for co=0...64
for ci=0...3
for r=0...3
for c=0...3
read()



// Conv2D

for r=0...3
for c=0...3
for ci=0...3
for h=0...34
...
if (r==2, c==2....)
write(FIFO)

② Delayed FIFO writes

